

Sales Prediction Proposal

Multiple Linear Regression

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I plan to predict Sales from TV, Radio, Social Media, and Influencer type. I will use the Dummy Marketing and Sales dataset by Harriman Samuel Saragih on Kaggle. It is a dummy dataset with 4,572 rows and five columns. TV, Radio, Social Media, and Sales are numbers, while Influencer has four categories.

I will check missing values, duplicate rows, data types, and category names. I will remove rows with missing values because only a small number of rows are affected. I will use scatter plots to study the link between each advertising variable and Sales, and a bar chart to compare average Sales by Influencer type.

I plan to use multiple linear regression because Sales is a numerical value and I want to use several predictors. I will change Influencer categories into 0/1 columns, then use 80% of the data for training and 20% for testing. I will check Mean Absolute Error, Root Mean Squared Error, and R-squared, and use prediction and residual plots to check the results.

Data source

Harriman Samuel Saragih. Dummy Marketing and Sales Data. Kaggle.

[Dummy Marketing and Sales Data on Kaggle](#)

